

ARCHITECTURE

BobaLink Browser Extension — Architecture

Revision: 1 **Last modified:** 2026-06-10T23:55:00Z **Scope:** BobaLink (extension/) — WXT + TypeScript Manifest-V3 cross-browser extension that detects magnet links and .torrent URLs on supported pages and forwards them to the Boba merge service on port 7187. **Authority:** the real source tree under extension/src/; docs/browser_extension/Status.md (Rev 7).

Anti-bluff (§11.4.6). Every component, edge, and message type below traces to a real module under extension/src/ (verified by reading the source). No invented components. Where a name is load-bearing it is given exactly as it appears in the code (isValidScanResult, MENU_SEND_GROUP, addMagnets, readSessionPassphrase, POST /api/v1/download, ...).

1. Data-flow diagram

The diagram renders on GitHub. In the exported HTML it remains a fenced code block (per scope note) — the textual edges are the load-bearing record.

```
flowchart TD
    subgraph PAGE["Web page (content-script context – matches from SITE_SELECTORS)"]
        DOM["Page DOM<br/>(magnet: links, .torrent URLs, bare text)"]
        ENTRY_C["entrypoints/content.ts<br/>(WXT thin wrapper · run_at document_idle)"]
        CONTENT["content/index.ts<br/>initContentScript()"]
        ORCH["scanner/orchestrator.ts<br/>ScannerOrchestrator (cross-scanner dedup by stable id)"]
        SCAN["scanner/{link-scanner,text-scanner}.ts<br/>+ base.ts · site-db.ts"]
        PARSE["parser/{magnet,torrent-file,bencode}.ts<br/>(SHA-1 infohash)"]
        HL["content/highlight.ts<br/>HighlightManager (DOM marking, toggleable)"]
    end

    subgraph SW["Background service worker (MV3)"]
```

```

ENTRY_B["entrypoints/background.ts<br/>(WXT thin
wrapper)"]
  BG["background/index.ts<br/>handleMessage() router +
initBackground()"]
  GUARD["isValidScanResult()<br/>(trust boundary shape-
guard)"]
  STATE["tabResults: Map<tabId, PageScanResult><br/>+
action badge + notifications"]
  TG["tabgroups/index.ts<br/>batchGroupTorrents /
dispatchGroupBatch"]
  CLIENT["api/boba-client.ts<br/>BobaClient (token-bucket +
retry)"]
  QUEUE["api/queue.ts<br/>OfflineQueue (FIFO, persisted,
dead-letter)"]
  HEALTH["api/health.ts<br/>probeHealth()"]
end

  subgraph CRED["Credentials (§11.4.10)"]
    CRYPTO["shared/crypto.ts<br/>AES-256-GCM decrypt() ·
PBKDF2"]
    SSTORE["chrome.storage.session<br/>session passphrase (in-
memory only)"]
  end

  subgraph UI["Extension UI pages"]
    POPUP["popup/popup.ts<br/>detected list + Send / Send-
All"]
    OPTIONS["options/options.ts<br/>servers, config, token
encrypt"]
    LSTORE["chrome.storage.local<br/>(shared/storage.ts)"]
  end

  BOBA["Boba merge service :7187 (FastAPI)<br/>POST /api/v1/
download · GET /health · GET /api/v1/auth/status"]

  DOM --> ENTRY_C --> CONTENT --> ORCH
  ORCH --> SCAN --> PARSE
  ORCH -->|scan-completed event| CONTENT
  ORCH -->|torrent-detected event| HL --> DOM
  CONTENT -->|chrome.runtime.sendMessage 'scan-result'| BG
  BG --> GUARD -->|valid| STATE

  POPUP -->|'get-detected'| BG
  BG -->|"{" result }"| POPUP
  POPUP -->|'send-torrent' ids| BG
  BG -->|MENU_SEND_GROUP| TG --> BG
  BG -->|addMagnet / addMagnets| CLIENT
  CLIENT -->|"POST /api/v1/download {result_id, download_urls}"|
BOBA
  CLIENT -.->|send failure → enqueueFailed| QUEUE
  QUEUE -.->|"processQueue(processQueueItem) retry"| CLIENT
  BG -->|'health-check'| HEALTH --> BOBA

  CLIENT -->|BobaClient.create decrypts token| CRYPTO

```

BG -->|readSessionPassphrase| SSTORE --> CRYPTO

OPTIONS -->|'get-config' / 'set-config'| BG

OPTIONS --> LSTORE

BG --> LSTORE

BG -->|"chrome.tabs.sendMessage 'scan-now' / 'highlight-toggle'"| CONTENT

Edge legend. Solid = primary request/response path; dotted = failure/retry path.
Quoted labels are real MessageType strings or method names from the code.

Narrative trace (page → backend)

1. **Page DOM → content script.** The WXT content entrypoint (entrypoints/content.ts, run_at: document_idle, matches derived from SITE_SELECTORS) calls `initContentScript()` (content/index.ts), which builds a `ScannerOrchestrator`.
2. **Orchestrator → scanners → parsers.** The orchestrator runs `LinkScanner` + `TextScanner` over the DOM, deduping by the `base.ts` stable id; the parsers (`magnet.ts`, `torrent-file.ts`, `bencode.ts`) extract the SHA-1 infohash. On each scan-completed event the content script sends a typed scan-result envelope up via `chrome.runtime.sendMessage`. The `HighlightManager` marks detected links in the DOM (toggleable).
3. **Background router.** `background/index.ts handleMessage()` receives scan-result, passes it through `isValidScanResult()` (the trust-boundary shape-guard — a non-array items is rejected so a hostile page cannot poison the set), and stores it in `tabResults` keyed by tab id, updating the action badge.
4. **Send.** On send-torrent (or `MENU_SEND_GROUP`), the background builds a `BobaClient`, decrypting the configured token, and calls `addMagnet / addMagnets`, which `POST /api/v1/download` to :7187 with `{result_id, download_urls: [...]}`.
5. **Failure → offline queue → retry.** Each failed send is `enqueueFailed`-ed into the persisted `OfflineQueue`; `processQueue(processQueueItem)` re-attempts queued items through a fresh client, dead-lettering after `maxRetries`.
6. **Token decrypt.** Building the client (`BobaClient.create`) decrypts the AES-256-GCM encrypted `BobaApiToken` bundle via `shared/crypto.decrypt`, keyed by the session passphrase read from `chrome.storage.session (readSessionPassphrase)`. Default-open when locked — the ciphertext is never sent.
7. **Popup / options / health.** The popup queries `get-detected` and dispatches `send-torrent`; the options page round-trips `config` via `get-config / set-config`; the background's health-check and a periodic alarm probe `GET / health` via `probeHealth`.

2. Module responsibilities

Module (extension/src/...)	Responsibility
parser/bencode.ts	Bencode decoder for .torrent file bodies.
parser/magnet.ts	

Module (extension/src/...)	Responsibility
	Parse magnet: URIs; sanitizeDisplayName (ReDoS-hardened linear tag-strip).
parser/torrent-file.ts	Parse .torrent files; compute the SHA-1 infohash.
scanner/base.ts	BaseScanner contract + computeStableId (infohash → magnet URI → file URL → name).
scanner/link-scanner.ts	Detect torrents in <a href> links.
scanner/text-scanner.ts	Detect bare-text magnets / .torrent URLs in page text.
scanner/orchestrator.ts	Run all scanners, cross-scanner dedup by stable id, MutationObserver re-scan; emit scan-completed / torrent-detected.
scanner/site-db.ts	Curated per-site selector table (single source of truth).
content/index.ts	initContentScript() — orchestrator + highlight + content[?]background messaging; sends scan-result.
content/highlight.ts	HighlightManager — DOM marking of detected links, runtime-toggable.
background/index.ts	MV3 message-hub: handleMessage router, context menus, keyboard commands, alarms, badge, notifications, send + enqueue; isValidScanResult trust-guard; MENU_SEND_GROUP.
popup/popup.ts	initPopup(doc) — detected-torrent list, per-row Send + Send-All, status indicator; talks to background ONLY via messages; safe DOM (no innerHTML).
options/options.ts	Servers/config UI, token encryption, WAI-ARIA tablist (Arrow/Home/End keyboard nav).
api/boba-client.ts	BobaClient — the one network client to :7187; addMagnet/addMagnets (POST /api/v1/download), health, authStatus; token-bucket rate-limit + exponential-backoff retry; BobaClient.create decrypt-and-send.
api/queue.ts	OfflineQueue — persisted FIFO retry queue; priority order, FIFO-evict at maxSize, retry-state, dead-letter; SEND injected as a QueueSender.
api/health.ts	probeHealth() — liveness/latency/version probe of a server.
tabgroups/index.ts	batchGroupTorrents / dispatchGroupBatch — dedupe a tab group, one batched addMagnets POST.
shared/crypto.ts	AES-256-GCM encrypt/decrypt, PBKDF2 key derivation, EncryptedBundle, isEncrypted.

Module (extension/src/...)	Responsibility
shared/storage.ts	storageGet/storageSet over chrome.storage.local.
shared/events.ts	TypedEventEmitter + EventMap (in-process domain events; scanner [?] content).
shared/logger.ts	createLogger/initLogger (token/passphrase never logged).
shared/constants.ts	SITE_SELECTORS, STORAGE_KEYS, BADGE_COLORS, BOBA_TOKEN_HEADERS, RATE_LIMIT, RETRY_CONFIG, REQUEST_TIMEOUTS, EXT, ENCRYPTION, DEBOUNCE_DELAYS.
shared/utils.ts	TokenBucket, debounce, sleep, truncate, yieldToBrowser.
shared/errors.ts	NetworkError, ServerError, StorageError.
types/ {api, config, torrent}.ts	MessageType / ExtensionMessage / PageScanResult / DetectedTorrent / ExtensionConfig / ServerConfig.

3. WXT entryptoint / thin-wrapper layout

WXT generates the MV3 manifest and bundles the worker/scripts from the entryptoints/ directory. Each entryptoint is a **thin wrapper** that calls into a fully unit-testable logic module, so the test corpus imports the logic modules directly while the entryptoints own the real browser lifecycle.

Entryptoint	Wraps	Manifest role
entryptoints/ background.ts	initBackground() (background/ index.ts)	background.service_worker. Idempotent — safe even if the logic module auto-registers on import.
entryptoints/ content.ts	initContentScript() (content/index.ts)	content_scripts with matches derived from SITE_SELECTORS (no <all_urls>), run_at: document_idle, all_frames: false. SINGLE driver of the content logic (the logic module's self-init was removed to avoid double-registering the message listener).
entryptoints/ popup/ index.html	loads popup/popup.ts	action.default_popup.
entryptoints/ options/ index.html	loads options/ options.ts	options_ui / options page.

Build output: loadable `.output/chrome-mv3/` + `.output/firefox-mv2/`, plus per-store zips `bobalink-1.0.0-{chrome,firefox,sources}.zip` (produced by `extension/ci-ext.sh`, §11.4.38 artifact-verified).

4. Message-type catalog

`MessageType` is the closed union in `types/api.ts`. The catalog below lists the messages actually routed by the two `handleMessage` switches (background in `background/index.ts`, content in `content/index.ts`) and their direction.

Sender → Background (routed by `background/index.ts`)

Type	Sender	Background action
scan-result	content script	Trust-guard via <code>isValidScanResult</code> ; store in <code>tabResults</code> ; update badge.
get-detected	popup	Return the tab's stored <code>PageScanResult</code> (<code>{ result }</code>).
send-torrent	popup	<code>sendTorrents(tabId, ids) → BobaClient.addMagnet</code> ; enqueue failures.
health-check	popup / options	<code>probeHealth</code> every configured server; return per-server status.
queue-status	popup / options	Return <code>OfflineQueue</code> size + items.
queue-process	popup / options	<code>offlineQueue.processQueue(processQueueItem)</code> (retry drain).
get-config	popup / options	Return the persisted <code>ExtensionConfig</code> .
set-config	options	Persist config; re-init the logger.
scan-page	popup / command	Forward scan-now to the tab's content script.
open-dashboard	popup	Open the active server URL in a new tab.

Background → Content (`chrome.tabs.sendMessage`, routed by `content/index.ts`)

Type	Content action
scan-now	Re-scan the page (<code>rescan() → orchestrator.scanNow</code>); re-sends scan-result.
highlight-toggle	Toggle (or set) <code>HighlightManager</code> DOM highlighting.
get-detected	Return the orchestrator's detected set. Flip a detected torrent's selected flag.

Type	Content action
toggle-selection	
get-scan-status	Report scanning/scanned state + count.

The unused-by-router members of the MessageType union (send-result, health-result, get-auth-state, authenticate, show-notification, update-badge, torrent-detected, selection-change) are declared response / reserved variants; the load-bearing routed set is the two tables above.

Sources verified 2026-06-10

- extension/src/background/index.ts — message router, isValidScanResult, MENU_SEND_GROUP, alarms, readSessionPassphrase, clientFor
- extension/src/content/index.ts — initContentScript, scan-result send, content message switch
- extension/src/api/boba-client.ts — POST /api/v1/download, addMagnet(s), BobaClient.create decrypt-and-send
- extension/src/api/queue.ts — OfflineQueue, injected QueueSender
- extension/src/scanner/orchestrator.ts — cross-scanner dedup
- extension/src/shared/{crypto,events,constants,storage}.ts
- extension/src/entrypoints/{background.ts,content.ts} — thin-wrapper layout
- extension/src/types/api.ts — MessageType union